

OIL INDIA LIMITED (A Govt. of India Enterprise)

DRILLING & WELL SERVICES DIRECTORATE - DULIAJAN - 786602, ASSAM
OPERATIONAL DIGEST & DAILY DRILLING REPORT (DDR)

REPORT NO: DDR-TGK-01/24

DATE: 18-JUN-2024

REPORT STATUS: VERIFIED

1. WELL IDENTIFICATION & RIG OPERATIONAL TELEMETRY

Well ID: OIL-TENGAKHAT-1	Current Depth TVD: 2,160.0 m	Rig: OIL Rig-E2000 (2000 HP)
Field: Tengakhata (Upper Assam)	Current Depth MD: 2,225.0 m	Mud Weight: 11.4 ppg (WBM)
Block: AA-ONN-2015/09	Active Formation: Barail Formation	Kelly Bushing (KB): +42.0 m MSL

2. DOWNHOLE HAZARDS, OPERATIONAL INCIDENTS & NPT REMEDIATION RECORDS

Incident 1: Gas Kick (Formation Influx & Abnormal Pore Pressure) SEVERITY: CRITICAL

Depth: 2145.0m TVD	Formation: Barail Formation	NPT: 6.5 hrs	Severity: CRITICAL
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Incident 2: Lost Circulation (Sub-hydrostatic Fractured Zone) SEVERITY: HIGH

Depth: 1480.0m TVD	Formation: Tipam Sandstone	NPT: 4.0 hrs	Severity: HIGH
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Root Cause: Sub-hydrostatic, high-permeability micro-fractured sandstone interval penetrated in Upper Tipam at 1,480m TVD. Formation fracture breakdown gradient lower than hydrostatic mud column, initiating dynamic partial mud loss rate of 42 bbl/hr into the formation.

3. ACTIVE DRILLING FLUID RHEOLOGY & HYDRAULICS MONITORING

Mud Weight (Density):	11.4 - 12.1 ppg	Active system weighted up with Barite post-kick kill operation
Plastic Viscosity (PV):	22 cP	Within API spec (18-28 cP) for efficient Barail cutting transport
Yield Point (YP):	19 lb/100ft ²	Optimized hole cleaning without excessive ECD surging
API Fluid Loss (FL):	4.8 ml/30min	PAC-LV polymer control preventing deep filtrate invasion
Equivalent Circulating Density (ECD):	12.35 ppg	Pore pressure margin: +0.25 ppg overbalance maintained safely

4. STRATIGRAPHIC HORIZON EVALUATION & TARGET TOPS

Tipam Sandstone Member	Top: 1,220m TVD	Encountered loss zone at 1,480m. Healed with 45 bbl LCM pill.
Girujan Clay Member	Top: 1,840m TVD	Massive impermeable mottled clay. Normal hole condition.
Barail Main Sandstone Member	Top: 2,120m TVD	Gas influx at 2,145m. Killed with 12.1 ppg mud. Drilling resumed.

5. REPORT SIGN-OFF & REGULATORY VERIFICATION

Prepared By: A. K. Gogoi (Lead Drilling Engineer)	Verified By: D. K. Baruah (Chief Geologist)
Designation: Rig Superintendent, OIL Rig-E2000	Subsurface Directorate, Oil India Limited, Duliajan